

Vector Search

Finding semantically similar information with embeddings

What is vector search?

Vector search retrieves information by comparing numerical representations of data rather than relying only on exact keyword matches. Text is transformed into an embedding vector, and search compares the query vector with stored document vectors to identify semantically similar content.

Embeddings

An embedding model maps text into a high-dimensional numerical space. Texts with related meaning tend to have representations that are closer together according to a similarity metric. Common choices include cosine similarity, dot product, and Euclidean distance, depending on the embedding model and search system.

Indexing

A vector database or vector index stores embeddings together with metadata and the original content or a reference to it. During ingestion, documents are usually split into chunks, embedded, and indexed. Metadata such as document name, section, timestamp, or source can be retained for filtering and traceability.

Search process

At query time, the user's question is embedded using the same embedding model. The vector index then finds the nearest stored vectors. The system can return the top-k matches, optionally apply metadata filters, rerank candidates, and pass the strongest results to a downstream application such as a RAG pipeline.

Trade-offs

Vector search is useful for semantic discovery but introduces engineering choices around chunk size, embedding quality, index type, latency, and recall. A good system balances retrieval quality with computational cost and often combines semantic retrieval with keyword or metadata filtering.

Reference document for the Veritas portfolio demonstration.